Project Title: Video Game Sales Analysis

Problem Statement:

The video game industry is a rapidly evolving sector, with sales performance varying across different platforms, genres, and regions. Analyzing game sales data can provide valuable insights into market trends, best-selling games, and revenue distributions. This project aims to explore a dataset containing sales data of video games to answer key business and analytical questions.

Objective:

The primary goal of this project is to analyze video game sales data to:

- Identify top-selling games globally and regionally.
- Understand the distribution of sales across different platforms and genres.
- Examine trends in sales over different years.
- Analyze how publishers influence sales performance.

Fields include:

Rank - Ranking of overall sales

Name - The games name

Platform - Platform of the games release (i.e. PC,PS4, etc.)

Year - Year of the game's release

Genre - Genre of the game

Publisher - Publisher of the game

NA_Sales - Sales in North America (in millions)

EU_Sales - Sales in Europe (in millions)

JP_Sales - Sales in Japan (in millions)

Other_Sales - Sales in the rest of the world (in millions)

Global_Sales - Total worldwide sales.

Tasks:

Top Selling Games:

- What are the top-selling games based on global sales revenue?
- What is the total global sales revenue across all years?

Platform Performance:

- Which platform has the highest total global sales revenue?
- Are there any platforms that consistently produce top-selling games over time?
- How does the distribution of global sales revenue vary across different platforms?

Genre Analysis:

- What are the most popular genres in terms of global sales revenue?
- Are certain genres more successful on specific platforms or in particular regions?
- Can we identify any emerging trends or shifts in genre preferences over time?

Publisher Performance:

- Which publisher has the highest total global sales revenue according to genre?
- How does the performance of publishers vary in different regions?

Regional Sales Analysis:

- How do sales revenues vary across different regions (e.g., North America, Europe, Japan, Other)?
- Can we identify any regional preferences or trends in gaming preferences?

Approach:

To address these questions, the following steps were taken:

Data Preparation:

- Use Pandas to inspect missing values **(df.isnull().sum())**.
- Impute missing data where feasible or drop rows with insufficient information.
- Convert **Year** column into datetime format using **pd.to_datetime()** if applicable.

Exploratory Data Analysis (EDA):

- Generate descriptive statistics using **df.describe()** for numerical columns.



- Use **groupby()** to calculate total and average sales by **region, platform**, and **genre**.
- Visualize distributions using Seaborn histograms and box plots.

Top Selling Games Analysis:

- Sort the dataset by **Global_Sales** to identify top-selling games.
- Calculate the total global sales revenue by summing **df['Global_Sales'].sum().**

Platform Performance Analysis:

- Group data by Platform and aggregate total sales using **df.groupby('Platform')['Global_Sales'].sum()**.
- Use bar plots to visualize platform-wise sales performance.

Genre Analysis:

- Aggregate data by **Genre** to determine high-performing categories using **df.groupby('Genre')['Global_Sales'].sum()**.
- Create bar charts to compare sales figures across different game genres.
- Analyze time trends by grouping sales data by **year** and **genre**.

Publisher Performance Analysis:

- Group data by **Publisher** and summed **Global_Sales** to identify top-performing publishers.
- Compare publisher performance across different regions using **groupby(['Publisher', 'Region'])['Sales'].sum()**.

Regional Sales Analysis:

- Aggregate data by **Region** to understand market contributions.
- Visualize regional sales trends using stacked bar charts.
- Identify region-wise preferences by analyzing top-selling games per region

